

Technical Architecture Overview

Companion to the One-Pager · For Kevin Packer / InvesTech Consulting · 2026-05-18

This is the diligence-grade companion. Every number below was primary-source verified from the codebase on 2026-05-17.

1. Codebase scale at a glance

Metric	Count	Verified via
PostgreSQL tables	271	grep of Drizzle schema
Frontend page files	211	find client/src/pages
Server route/logic files	84	find server -maxdepth 2
Production line-count in 10 highest-density pages	~27,000 lines	explorer audit
Live trade simulations	6 × 15 lessons = 90	tradeSimsLessons table
AI engines in synthesis	4	server/ai-provider.ts
RAG knowledge chunks	86	server/rag-engine.ts
Public service platforms operated	15	ecosystem catalog
Grants tracked	651	grant_opportunities table
Languages (Trade Sims AI tutor)	10	client/src/lib/i18n
Talk Your Talk language coverage	107 (89 spoken + 18 signed)	TYT registry

Stack: **React + Vite + TypeScript** (frontend) · **Express + Node.js** (API) · **PostgreSQL (Neon)** via **Drizzle ORM** · **Replit Auth (OIDC)** · **TanStack Query v5** · **wouter routing** · **Tailwind + shadcn/ui**.

2. The four physics engines behind Trade Sims

These are not "real physics" generically. They are industry-standard solvers running in the browser.

Trade	Engine	Standard
Electrical	Modified Nodal Analysis (MNA) DC solver	Standard EE undergraduate solver
Automotive	MNA via adapter (reuses electrical solver)	Engineering-clean reuse pattern
Plumbing	Hardy-Cross Newton-Raphson flow solver	Standard civil/mechanical pipe-network solver, with active-set outer loop for true one-way check valves
Welding	Heat-Input Evaluator	AWS D1.1 §5.7 (American Welding Society Structural Welding Code)
HVAC	Thermal-airflow engine	Production 2026-05-17
Software Engineering	Concept-only (no physics)	15 lessons covering algorithms, version control, testing, design patterns, scalability, code quality, security

Test coverage (self-running, exits 1 on failure):
Heat-input-evaluator: 16/16
Plumbing flow-solver: 17/17 (including parallel-branch and reverse-flow test cases)
Automotive thermal-evaluator: 10/10

integrated LLM passing (KVL/ KCL, voltage-divider, series-parallel sanity)
Credentialed routing at 80% lesson completion. Production at server / trade-sims-credentials.ts per trade (OSHA 10, NCCER L1, AWS SENSE, AWS D1.1, ASE G1, EPA 609, EPA 608 Universal, NATE Ready-to-Work, GitHub Foundations, CompTIA ITF+, AWS CCP, AZ-900, (ISC)² CC, LFCA, state apprenticeship pathways per trade (IBEW/NECA, UA Local 286, ABC, Iron Workers, SMART, OEM tech programs, Apprenti, Microsoft LEAP, Multiverse, Year Up, AWS re/Start, TDR, TSPC)
+ 26 registered apprenticeship on every cert (Iron Rule)

3. AI stack — orchestration, not wrapping

Capability	Where it lives	Why it matters
4-engine collaborative synthesis	server/ai-provider.ts	Any one provider outage does not break the session
86-chunk RAG	server/rag-engine.ts	Grounded in TCAF's own active-commitment docs
Trade Sims AI tutor — two modes	server/trade-sims-routes.ts	Socratic-hint mode (no answer-giving) + ensemble-debrief mode
Foster-youth AI risk engine	server/foster-youth-risk.ts	4-domain scoring: housing / food / mental health / documents
AI early-warning engine	server/early-warning.ts	Triggers intervention playbooks
AI proposal/narrative generator	server/loi-routes.ts, grant-packages.tsx	50-state-compliant narrative generation
AI grant fit-scoring	server/grant-routes.ts	Tier-weighted keyword scoring + semantic analysis
AI translation — dialect-aware	server/translate-routes.ts	Preserves AAVE, Spanglish, regional dialect, not just literal translation
Deterministic agent-knowledge bootstrap	GET /api/agent/knowledge/session-bootstrap	~3KB session-startup memory hook

Funder framing: "We don't use AI. We orchestrate it." Four engines, failover, mode-switching tutors, dialect-aware translation, RAG grounded in our own documented commitments.

4. Implementation Science — code, not slogans

- 39 CFIR constructs wired into client/src/pages/research-

• **NRC + CFIR 2.0 fidelity scoring rubrics** in server/standard-cqi-client/src/pages/map-gap-cqi.tsx is 1,705 lines of continuous-improvement log-outcome-reporting-detection grade evidence dashboards in ecosystem-rplce-bridge.ts auto-routes implementation-science findings to relevant platform domains

5. Justice & Reentry stack

11 production tables. Not pitched in most prior materials

- **RNR Assessments** — Risk-Need-Responsivity, the gold standard in correctional programs
- **CBT Programs** — Cognitive Behavioral Therapy
- **Recidivism baselines** — tracked long-term outcomes
- **Family visitation tracking** — NRC could come reports
- **NRC Co-located reports** — National Reentry Resource Center standard

standards in justice-command-center.tsx — 3,482 lines of integrated case management

6. Foster-Youth Transition Engine

- 50-state policy comparator with statute citations (client/src/data/foster-youth/state-policy-charts), Medicaid-to-26, John H. Chafee Foster Care Independence, state-specific eligibility requirements on the public no-auth wizard: server-issued per-row accessToken, x-intake-token header, timingSafeEqual comparison, per-IP rate limits on AI/ML model health-check generator for caseworkers

7. Compliance & Security posture

- **FHIR + CDS Hooks** clinical interoperability (SafeReport, live at safereports.net — "Compliance-Goal-Adherence-Goal-Get-pager")
- **PHQ-9 longitudinal screening instruments**: PHQ-9, GAD-7, C-SSRS system, auth-x-ecosystem-key + x-shadow-key server-side exact-match comparison (no prefix-checking)
- **Adding logging**: academy_choice_logs + threat_model_published at threat_model.md covering

spoofing, tampering, repudiation, information disclosure, DoS, and elevated identity capability strategy: TCAF (501c3) + ISS LLC (for-profit) — clean firewall

8. Grant Intelligence Engine

• **651 grants tracked** as of 2026-05-17 (grants.gov (369) · USASpending.gov (198) · SAM.gov (36) · manual (12) · state/local (18) · other federal (8) · foundation/corporate (4) · **weighted keyword scoring** with explicit point values (e.g., "PHI-safety" = 100, "disaster" = 50, "manual" = 20, "cron scheduled": GET /api/grants/this-week, POST /api/grants/this-week: strategic dimensions, ship targets, grants closing in 14 days, **6,340 lines** of multi-document assembly + narrative engine — **3,182 lines** of real-time AI scoring + pipeline ops
SAM.gov honest framing: we screen the 16,000+ active-opportunity feed and curate ~36 of direct relevance. We never claim to "track 16,667."

9. Academy Live Economic Engine — almost never surfaced

45 Academy Live, including: academyStocks, academyPortfolios — live economic simulation for student leaders, academyCompetitions — gamified math orders — real funding flow, academyPower — GAM-ready (Generalized Additive Model) scenarios + nodes + logs — branching interactive narratives, academyLifeLessons — financial literacy, civic engagement, social navigation, academyBehavioralAuditTrail — This is FAFSA + financial literacy + AI literacy + STEM + workforce + apprenticeship + behavioral-economics simulator under one auth.

10. Ecosystem Hub — real cross-platform

architecture

- **Heartbeat system:** 25 platforms tracked live (15 public, 10 internal/
- **Cross-platform event bus:** deduplication
- **Shadow Observer pattern:** server/agent-communication.ts for AI
- **APLICE-Bridge flow:** implementation-science findings auto-routed to relevant platform
- **Evidence Registry:** communityEvidence table citing
- **Platform-Funder fit table:** formal data structure for matching platforms to funders

11. Build-heavy pages (illustrative)

Page	Lines	What it is
grant-packages.tsx	6,340	Multi-doc assembly + narrative engine
justice-command-center.tsx	3,482	RNR-integrated case management + recidivism dashboard
grant-command-center.tsx	3,182	Real-time AI grant scoring + pipeline ops
rplice-tools.tsx	2,750	Implementation-science scoring + community analysis
presentations.tsx	1,970	Integrated slide generation + stakeholder pitch engine
opportunity-youth.tsx	1,930	High-density persona-based pathway modeling
map-gap-cqi.tsx	1,705	CQI logic for ecosystem gaps
wab2-enrollment-hub.tsx	1,597	Multi-factor eligibility + enrollment engine
transition-plans.tsx	1,547	Collaborative plan builder with milestones
apprenticeship-tracker.tsx	1,538	Longitudinal trade-placement tracking

~27,000 lines in those 10 pages alone. Shipped, in production, with real users.

12. Iron Rule

Every number in this document was primary-source verified from the codebase on 2026-05-17. Any number you want to cite externally should be re-verified against the funder's diligence date — memory is never a primary source — Terry D. Flood Sr., President · TCAF · 2026-05-18